13.3 The dot product

The dot product of $\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$ and $\vec{w} = w_1\vec{i} + w_2\vec{j} + w_3\vec{k}$ is

$$\vec{v}\cdot\vec{w} = \underbrace{v_1w_1 + v_2w_2 + v_3w_3}_{\text{this a number, not a vector!}} \quad \leftarrow \text{"algebraic definition"}$$

Equivalently, we also have

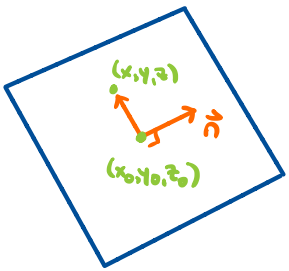
$$\vec{v}\cdot\vec{w} = \|\vec{v}\|\,\|\vec{w}\|\cos(\theta) \quad \leftarrow \text{"geometric definition"}$$

where θ is the angle $(0 \le \theta \le \pi)$ between $\vec{v}$ and $\vec{w}$.

Facts

- $\vec{v}$ and $\vec{w}$ are perpendicular, or orthogonal, if and only if $\vec{v}\cdot\vec{w} = 0$
- $\vec{v}\cdot\vec{v} = \|\vec{v}\|^2$.

→ Do problems 1–3.



Let $\vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$ be a vector in $\mathbb{R}^3$ and $P = (x_0, y_0, z_0)$ be a point. The equation of the plane through P and with normal vector $\vec{n}$ is

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

→ Do problems 4 and 5.